Foundations of Machine Learning

A Study Guide to Supervised Learning | Regression & Classification

1. Introduction to Machine Learning

The Samuel Definition

In the 1950s, Arthur Samuel defined machine learning as the branch of study that gives computers the ability to learn without being explicitly programmed. This was a paradigm shift: traditional programming needs a human to hand-write instructions for every scenario, while machine learning lets a system gain experience and improve on its own.

The Checkers Case Study

Samuel was not a strong checkers player himself, so he had the program play tens of thousands of games against itself. By tracking which positions led to wins versus losses, the program learned to recognize "good" and "bad" positions - and eventually became better than its own creator. This showed a machine can exceed the skill level it was initially given, purely by finding patterns in data.

The Data Volume Principle

There is a direct cause-and-effect link between data volume and accuracy: fewer games played would have meant weaker performance. More experience generally leads to better performance.

> **Why this matters for you:**
>
> This is the seed idea behind everything that follows: models get better with more (good) data, not with cleverer hard-coded rules.

2. The Architecture of Supervised Learning

The Core Mechanism: X to Y Mapping

Supervised learning powers most modern AI applications. Its job is to learn an input-to-output ($x \rightarrow y$) mapping: take an input x and produce an accurate prediction of output y.

The Role of "Right Answers"

"Supervised" refers to training on a dataset that already contains correct labels (y) paired with inputs (x). By seeing thousands or millions of these correct pairs, the algorithm works out the underlying relationship

between input and output.

Generalisation: The Ultimate Goal

It is not memorisation
It's just you train model & if something he has never seen before he still produce right answer

A good model does not simply memorise the training data. It learns logic general enough to take a brand-new input it has never seen and still produce an accurate output. Predicting well on unseen data is the entire point of supervised learning.

3. Regression: Predicting Continuous Values

Regression predicts a number from infinitely many possible values - used when the target is a continuous scale rather than a fixed set of labels.

> **Worked Example - Housing Price Prediction**
>
> Input (x): size of the house in square feet.
> Output (y): price of the house in thousands of dollars.
>
> The algorithm fits a function (a line or a curve) to existing data points, then uses that function to estimate prices for houses not in the dataset. For a 750 sq ft house, a straight-line fit might predict $150,000, while a curve-fit might predict closer to $200,000. The key insight: the algorithm itself chooses the most appropriate function shape - that choice is part of what "learning" means here.

4. Classification: Categorical Prediction

Classification predicts a small, finite set of discrete categories ("classes"), which can be non-numeric (cat vs. dog) or represented as discrete numbers (0 or 1).

> **Worked Example - Breast Cancer Diagnosis**
>
> Input (x): tumor size.
> Output (y): 0 (benign) or 1 (malignant).
>
> Because the goal is one of two fixed categories rather than any number on a scale, this is classification, not regression.

Scaling Complexity with Multiple Inputs

Real systems often use multiple inputs at once - e.g. patient age AND tumor size - and the algorithm fits a boundary line to separate the categories. Clinical-grade models add more inputs still, such as uniformity of cell size, uniformity of cell shape, and clump thickness, to help find the best boundary.

5. Regression vs. Classification

Feature	Regression	Classification
Output Type	Continuous numbers	Discrete categories / classes
Possible Values	Infinitely many (e.g. any price)	Small, finite set (e.g. 0, 1, 2)
Primary Visualisation	Best-fit lines or curves	Boundary lines separating classes

Misconception Alert

Classification numbers like 0 and 1 are labels, not quantities. Regression can predict any value (0.5, 1.7, 183,000). Classification only ever predicts from the fixed set of categories defined up front - the numbers are just names for those categories.

6. Economic Impact and Real-World Applications

Supervised learning is responsible for roughly 99% of the economic value generated by AI today. Its ability to automate high-stakes predictions and categorisations makes it the primary driver of AI innovation across industries.

- **Online Advertising** - user + ad info → probability of a click (the most lucrative application).
- **Spam Filtering** - email content → "Spam" or "Not Spam".
- **Speech Recognition** - audio clip → accurate text transcript.
- **Machine Translation** - text in one language → translation in another (Spanish, Arabic, Hindi, Chinese, Japanese, etc.).
- **Self-Driving Cars** - camera + radar data → precise positions of surrounding objects/cars.
- **Visual Inspection (Manufacturing)** - product images → "Defect" or "No Defect".

7. Practical Implementation and Best Practices

The Tool vs. Skill Analogy

Having ML algorithms available is like being handed a top-of-the-line hammer or drill - powerful, but it won't build a three-story house by itself. Real success needs both the tools and the skill to apply them well.

The Cost of Inexperience

Even experienced ML teams at top tech companies have spent six months on a problem and failed because they used the wrong approach. Following established best practices is what prevents these costly detours.

8. Conclusion and Future Pathways

Supervised learning is the most-used branch of machine learning, valued for mapping inputs directly to "right answers." It sits inside a broader ecosystem of learning methods built for other kinds of problems:

- **Unsupervised Learning** - finding hidden patterns in unlabeled data.
- **Recommender Systems** - personalising suggestions for products or content.
- **Reinforcement Learning** - training agents to improve via rewards.

Study guide synthesised from lecture transcripts - Course 1: Supervised Machine Learning: Regression and Classification.